

School of Physics

Data Processing and Analysis Guide for Stage 6 Physics

Table of physical constants

Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Radius of the Earth	$6.371 \times 10^6 \text{ m}$
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$
	931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Absolute zero (0 K)	-273.15°C
Wein's displacement constant, b	$2.898 \times 10^{-3} \text{ m K}$

Introduction

This guide is intended for teachers of the Stage 6 Physics course in New South Wales. It draws upon resources published by the NSW Department of Education in their *Working Scientifically* support documents as well as those from the School of Physics First Year Unit at the University of New South Wales, Sydney.

The definitions and procedures described in this guide take into consideration the differences between what is done in tertiary studies and research in physics, and the skills and understanding of students studying physics in high school.

It is intended to give teachers and their students a sense of good practice for processing and analysing data. It is in no way designed to be prescriptive; teachers should consider the learning requirements and outcomes for their students when using this guide.

References to the NSW Physics Stage 6 Syllabus

PH11/12-4 Processing data and information

A student selects and processes appropriate qualitative and quantitative data and information using a range of appropriate media.

Students:

- select qualitative and quantitative data and information and represent them using a range of formats, digital technologies and appropriate media (ACSPH004, ACSPH007, ACSPH064, ACSPH101)  
- apply quantitative processes where appropriate 
- evaluate and improve the quality of data  

PH11/12-5 Analysing data and information

A student analyses and evaluates primary and secondary data and information

Students:

- derive trends, patterns and relationships in data and information
- assess error, uncertainty and limitations in data (ACSPH004, ACSPH005, ACSPH033, ACSPH099) 
- assess the relevance, accuracy, validity and reliability of primary and secondary data and suggest improvements to investigations (ACSPH005)  

Suggestions and corrections

Please forward any suggestions or corrections by email to schools@phys.unsw.edu.au

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Contents

1 – Recording data	4
1.1 Reporting data	4
1.1.1 <i>Scientific notation and orders of magnitude</i>	4
1.1.2 <i>Significant figures</i>	4
1.1.3 <i>Units</i>	5
1.2 Tabulating data	6
2 – Errors and uncertainties	7
2.1 Accuracy, errors and uncertainties	7
2.1.1 <i>Accuracy and error</i>	7
2.1.2 <i>Accuracy and uncertainty</i>	7
2.2 Quantifying uncertainty	7
2.2.1 <i>Reporting uncertainties</i>	8
2.2.2 <i>Uncertainty in values read directly from a measuring device</i>	8
2.2.3 <i>Uncertainties due to environmental or human factors</i>	9
2.2.4 <i>Uncertainty in the average from a series of repeated measurements</i>	9
2.2.5 <i>Uncertainty in the slope of a line of best fit</i>	10
2.2.6 <i>Combining uncertainties</i>	11
3 – Analysing data	12
3.1 General principles	12
3.2 Linearized equations	13
3.3 Plotting graphs by hand	14
4 – Using spreadsheets	16
4.1 Entering data	16
4.2 Performing calculations	16
4.3 Plotting a graph	18
4.4 Obtaining the uncertainty in the slope	20
4.5 Common spreadsheet symbols and functions	21
4.5.1 <i>Calling on cells in functions</i>	21
4.5.2 <i>Order of operations</i>	21
4.5.3 <i>Entering data</i>	21
4.5.4 <i>Common functions</i>	22
Exercises	23
Solutions to exercises	27
References	27

1 – Recording data

1.1 Reporting data

1.1.1 Scientific notation and orders of magnitude

A neat way of recording very large or very small numbers is by using *scientific notation*. This is where numbers are written as a product of powers of ten, also called *orders of magnitude*. For example:–

- 2 560 000 m can be written as 2.56×10^6 m
- 0.0000183 m can be written as 1.83×10^{-5} m

These numbers are 11 orders of magnitude apart, since there are 11 powers of ten between 10^6 and 10^{-5} .

Note that numbers like 8.4×10^6 m have an order of magnitude of 7 because it *rounds up* to 10^7 .

1.1.2 Significant figures

Significant figures are important because they signal the accuracy or uncertainty in a value. The last significant figure in a number suggests that it is accurate to $\pm \frac{1}{2}$ of its place. For example, 5.3 km implies a distance with uncertainty of 5.30 ± 0.05 km but 5.34 km implies 5.340 ± 0.005 km.

In a measurement, non-zero digits are significant. Trailing zeroes of a whole number are generally not significant. These are used to keep the other figures in their correct places. Examples:–

- 27000 m – 2 and 7 are significant but not the zeroes. There are 2 significant figures.
- 27410 m – 2, 7, 4 and 1 are significant. There are 4 significant figures. This measurement is considered less uncertain than the first.

Note: It can sometimes be ambiguous as to whether a trailing zero is significant or not. For example, depending on the last significant zero, a value of 100 m could have been rounded to:

- the nearest hundred (1 sig fig), that is, from a value between 50 m and 150 m
- the nearest ten (2 sig fig), that is, from a value between 95 m and 105 m
- the nearest integer (3 sig fig), that is, from a value between 99.5 m and 100.5 m

Notice the decreasing uncertainty as the number of significant figures is increased.

There are some conventions to indicate the last significant zero, although these are not consistently applied:

- Explicitly stating the number of significant figures, such as “3400 (3 sf)”
- An overline or underline with the last significant zero – in 530 $\overline{0}$ and 5300 there are 4 significant figures.
- A decimal point with no trailing zeroes indicates all preceding zeroes are significant – in 23000. there are 5 significant figures.

The leading zeroes in a decimal are not significant. They are used to keep the other figures in their correct places. Example:–

- 0.0000672 kg – only 6, 7 and 2 are significant. There are 3 significant figures.

The zeroes between the figures are significant. Example:–

- 40.07 s – each figure is significant. There are 4 significant figures.

The last zero in a decimal is significant. Examples:–

- 4.70 L – each figure is significant. There are 3 significant figures.
- 0.30 J – 3 and last zero are significant. There are 2 significant figures.

The result of a calculation is only as accurate as the least accurate value used to compute it. So, when reporting the result of a calculation, the result must be rounded to the same as the smallest number of significant figures of any value used in the calculation. For example,

$$\text{Energy} = 3.457 \text{ W} \times 5.60 \text{ s} = 19.3292 \text{ J}$$

the answer can only be reported as 19.3 J because the smallest number of significant figures in the calculation was three.

Note: In calculations consisting of simple additions and subtractions *only*, answers should be given with the same number of *decimal places* as the term with the least number of decimal places in the calculation. For example:

$$\Delta T = 137.45^\circ\text{C} - 37.8766^\circ\text{C} = 99.5734^\circ\text{C}$$

The accuracy implied by the additional decimal places in one of the numbers is meaningless if the other number is more uncertain.

1.1.3 Units

Most numbers used in physics have associated units (exceptions are ratios and some constants). *Système international d'unités* (SI) and other metric units should be used, but occasionally non-SI units have their places.

Length – metres (m)
Mass – kilograms (kg)
Time – seconds (s)
Force – newtons (N)
Energy – joules (J)

Potential difference – volts (V)
Current – amperes (A)
Charge – coulombs (C)
Magnetic field strength – tesla (T)
Temperature – Kelvin (K), degrees Celsius (°C)

Prefixes are sometimes used to convert units into more convenient orders of magnitude. Some examples (and their multipliers, in scientific notation):–

Prefix	Symbol	Multiplier	Example
nano–	n	$\times 10^{-9}$	<i>nanometre (nm)</i>
micro–	μ	$\times 10^{-6}$	<i>microampere (μA)</i>
milli–	m	$\times 10^{-3}$	<i>millivolt (mV)</i>
centi–	c	$\times 10^{-2}$	<i>centigram (cg)</i>
deci–	d	$\times 10^{-1}$	<i>decilitre (dL)</i>
deca–	da	$\times 10^1$	<i>decade</i>
hecto–	h	$\times 10^2$	<i>hectopascal (hPa)</i>
kilo–	k	$\times 10^3$	<i>kilometre (km)</i>
mega–	M	$\times 10^6$	<i>megajoule (MJ)</i>
giga–	G	$\times 10^9$	<i>gigawatt (GW)</i>

Some units are *derived* from the SI base units. For example, velocity is measured in units of *metres per second* (m/s, also written as m s^{-1}). Derived units can be determined from the equations used to calculate its quantity—they follow the usual rules for multiplication and division. For example,

$$\text{gravitational field strength, } g = \frac{19.6 \text{ N}}{2.0 \text{ kg}} = 9.8 \frac{\text{N}}{\text{kg}} = 9.8 \text{ N kg}^{-1}$$

$$\text{momentum, } p = mv = 7 \text{ kg} \times 3 \text{ m s}^{-1} = 21 \text{ kg m s}^{-1}$$

$$\text{electric field strength, } E = \frac{V}{d} = \frac{100 \text{ V}}{0.08 \text{ m}} = 1250 \frac{\text{V}}{\text{m}} = 1250 \text{ V m}^{-1}$$

1.2 Tabulating data

Data should be recorded clearly in a table. Additional columns should be added for any processed data.

Best practice for a table in HSC Physics should include:

- a descriptive title or caption
- columns that have headings that include units, where appropriate; units are not included in the body of the table
- the independent variable, usually towards the left, and the dependent variable, towards the right
- figures aligned by decimal point
- uncertainties, if appropriate.

Example:

Table: the variation in the period of a pendulum as its length increases

	Ten Periods, $10T$ (s)					
Length (m)	Trial 1	Trial 2	Trial 3	Average	$\pm$ Uncertainty	T^2 (s^2)
0.30	11.25	11.70	11.76	11.57	0.26	1.34
0.40	13.56	13.05	13.06	13.22	0.26	1.75
0.50	14.34	14.30	14.53	14.39	0.11	2.07
0.60	16.35	15.93	16.23	16.17	0.21	2.61
0.70	17.23	17.04	17.74	17.33	0.35	3.00
0.80	19.63	19.41	19.24	19.43	0.20	3.77
0.90	20.47	20.73	20.19	20.47	0.27	4.19
1.00	21.20	20.33	22.01	21.78	0.84	4.49
1.10	21.15	22.12	22.09	21.78	0.49	4.75
1.20	23.33	23.92	23.30	23.52	0.31	5.53

2 – Errors and uncertainties

2.1 Accuracy

2.1.1 Accuracy and error

Accuracy refers to the closeness of an observation to its theoretical or actual value.

The term *error* is used to quantify the difference between a measured value and its “true” value. It does not mean mistake or blunder, as is its common usage.

According to one view of the world, there exists a “true” value for any measurable quantity. However, any attempt to measure the true value will give an observed value that inherently includes some “error” because of the very real human and physical limitations in making measurements.

The following equations can be used to quantify error and accuracy:

$$\text{Absolute error} = |\text{true value} - \text{observed value}|$$

$$\text{Relative error} = \frac{|\text{true value} - \text{observed value}|}{\text{true value}}$$

$$\text{Accuracy} = 100\% - \text{relative error}$$

Relative error and accuracy are usually expressed as a percentage.

For example, the “true” value of gravitational field strength on Earth’s surface is 9.8 N kg^{-1} . In an experiment carried out by a student, the value observed was 8.5 N kg^{-1} .

$$\text{Absolute error} = |9.8 \text{ N kg}^{-1} - 8.5 \text{ N kg}^{-1}| = 1.3 \text{ N kg}^{-1}$$

$$\text{Relative error} = \frac{|9.8 \text{ N kg}^{-1} - 8.5 \text{ N kg}^{-1}|}{9.8 \text{ N kg}^{-1}} = 13\%$$

$$\text{Accuracy} = 100\% - 13\% = 87\%$$

2.1.2 Accuracy and uncertainty

In this guide, the concept of *uncertainty* will be used instead of error, where uncertainty is the lack of certainty in an observed or estimated value. This concept is considered more appropriate in experimental physics because of the epistemological questions over what a “true” value really is. We will replace the idea of a “true” value with the idea of a “best estimate” – in practice, an average of many repeated measurements.

Uncertainty is usually reported with its measurement or estimate, for example, $4.6 \pm 0.4 \text{ kg}$.

An accurate measurement is one where the uncertainty is small. Additionally, if you are trying to measure a known value, the measurement is accurate if that known value lies inside the uncertainty bounds of the observation. For example, the gravitational field strength at Earth’s surface measured by experiment as $9.6 \pm 0.5 \text{ N kg}^{-1}$ agrees with the accepted value (9.8 N kg^{-1} lies between 9.1 and 10.1 N kg^{-1}).

2.2 Quantifying uncertainty

Any value that has been observed or estimated has an associated uncertainty. Values that have been recorded should be quoted with an associated uncertainty.

Absolute uncertainty is expressed in the same dimensions as the value. For example, 5.4 ± 0.2 m could be a length where the true value might lie 0.2 m above or below the estimation of 5.4 m.

Relative uncertainty is the size of the uncertainty as a fraction of the observed or estimated value and is usually presented as a percentage.

$$\text{Relative uncertainty} = \frac{\text{absolute uncertainty}}{\text{observed or estimated value}}$$

For example, for the estimation 5.4 ± 0.2 m, the relative uncertainty is $\frac{0.2}{5.4} = 3.7\%$

2.2.1 Reporting uncertainties

Uncertainty is almost always quoted as $x \pm \Delta x$ where x is the observed or estimated value and Δx is the uncertainty in absolute terms (not percentage, although this is sometimes acceptable). Uncertainty should be given to one significant figure; the observed or estimated value should be rounded to same number of decimal places as the uncertainty (any more decimal places than the uncertainty would be meaningless).

For example, the height of a building might be 14.7 ± 0.5 m, not 14.691 ± 0.53 m.

2.2.2 Uncertainty in values read directly from a measuring device

The uncertainty in an observation made directly from a measuring device is equal to half of the smallest readable graduation on the scale of the device:

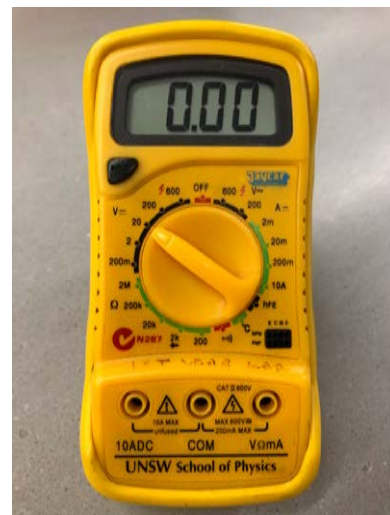
$$\text{Uncertainty} = \pm \frac{1}{2} \times (\text{smallest readable graduation on the scale of the device})$$

For example:



Analogue devices

This ammeter has graduations in milliamperes. Any observation made with this device will have an uncertainty of ± 0.5 mA.



Digital devices

This digital multimeter in 20VDC mode measures in increments of 0.01 V. Any observation made with this device in this mode will have an uncertainty of ± 0.005 V.

Sometimes the smallest graduation on a scale is not necessarily *readable*. For example, you may be using a metre rule marked in millimetres to measure the drop height of a ball, but as you conduct the experiment, you may only be able to measure to the nearest 5 centimetres. In this case, your measurements would have an uncertainty of ± 2.5 cm.

2.2.3 Uncertainties due to environmental or human factors

Some uncertainties may be due to the physical conditions or idiosyncrasies in the equipment you are using, for example: parallax, unstable oscillations, or noise.

Some ways to estimate and quantify these uncertainties could be:

- in the case of an unstable, fluctuating reading from a device – make a measurement of the highest value and the lowest value it fluctuates between. Determine the range between these values and divide it by two.
- in the case of parallax – estimate the maximum probable observation you could make and the minimum probable observation, find the range between them and divide by two.

For example, a light meter reading fluctuates between 979 lux and 1057 lux. The measurement would be halfway between the two numbers (1020 lux) and the uncertainty would be

$$\Delta I = \pm \frac{1057 - 979}{2} = \pm 39 \text{ lux}$$

It would be reported as 1020 ± 40 lux.

2.2.4 Uncertainty in the average from a series of repeated measurements

When an average is calculated from repeated observations in a series of trials, the uncertainty in the average can be calculated as half the range between the highest and lowest measurement:

$$\text{Uncertainty} = \pm \frac{\text{maximum value} - \text{minimum value}}{2}$$

For example, in the series of observations 21.20 s, 20.33 s and 22.01 s, the average and its uncertainty is 21.8 ± 0.2 s (remember, the uncertainty should be given to one significant figure, and the average should be to the same decimal place.).

This method is appropriate for when the number of repeated trials is small (i.e. 3 to 5 trials). More sophisticated techniques to calculate uncertainty can be used if the number of trials is sufficiently large (e.g. standard deviations).

2.2.5 Uncertainty in the slope of a line of best fit

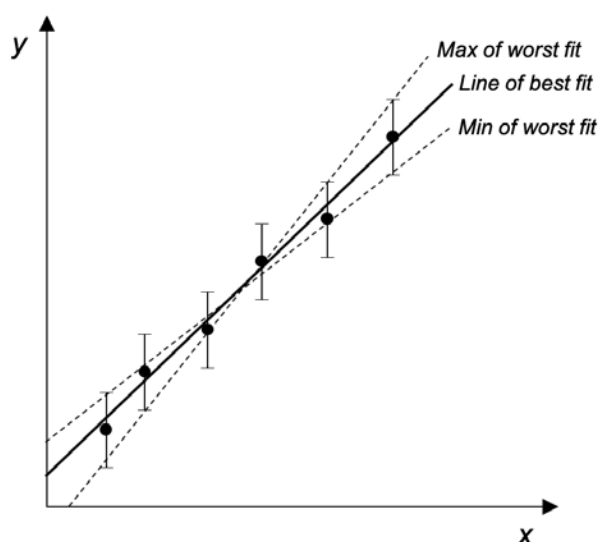
When a value has been determined using a line of best fit (linear regression), there is uncertainty associated with the gradient of the line.

To estimate its uncertainty, the data points should be plotted with their respective error bars (the length of an error bar is the uncertainty in each data point).

Two *lines of worst fit* should then be drawn. One line of worst fit is the *shallowest* straight line that can be drawn yet passing through the error bars of as many data points as possible. Similarly, the other line of worst fit is the *steepest* straight line that can be drawn.

The uncertainty in the slope of a line of best fit is then half of the difference between the gradients of the maximum and minimum lines of worst fit.

$$\text{Uncertainty} = \pm \frac{1}{2} \times (\text{maximum worst fit gradient} - \text{minimum worst fit gradient})$$



You should find that in general, data points that fit more closely to the line result in an uncertainty that is small. If the data is more spread out from the line, then the uncertainty is larger.

2.2.6 Combining uncertainties

When uncertain quantities are used in calculations, then the results are more uncertain.

–Uncertainties when values are added or subtracted

When adding or subtracting values that are uncertain, then their *absolute uncertainties* are added.

Case	Method
$a = b + c + \dots$ or $a = b - c - \dots$	$\Delta a = \Delta b + \Delta c + \dots$
$a = 2b$ (which is the same as $a = b + b$)	$\Delta a = 2\Delta b$

Examples:

- If a cable of length 7.2 ± 0.5 m is joined to another cable 5.6 ± 0.4 m, the length of the new cable would be 12.8 ± 0.9 m.
- If two masses were hung on a string, each 50.0 ± 0.5 g, then the combined mass would be 100.0 ± 1.0 g

–Uncertainties when values are multiplied or divided

When multiplying or dividing values that are uncertain, then the *relative uncertainties* are added.

Case	Method
$a = b \times c \times \dots$ or $a = b \div c \div \dots$	$\frac{\Delta a}{a} = \left(\frac{\Delta b}{b} + \frac{\Delta c}{c} + \dots \right)$
$a = b^2$ (which is the same as $a = b \times b$)	$\frac{\Delta a}{a} = \left(\frac{2\Delta b}{b} \right)$

Examples:

- When finding the area of a rectangle, given its length L is 5.2 ± 0.1 m and its width W is 7.9 ± 0.2 m,
 $A = L \times W = 5.2 \text{ m} \times 7.9 \text{ m} = 41.1 \text{ m}^2$
and its uncertainty

$$\Delta A = A \times \left(\frac{\Delta L}{L} + \frac{\Delta W}{W} \right) = 41.1 \text{ m}^2 \times \left(\frac{0.1}{5.2} + \frac{0.2}{7.9} \right) = 1.8 \text{ m}^2$$

so the result is $41 \pm 2 \text{ m}^2$.

- Given that the period of a pendulum $T = 2.35 \pm 0.03$ s, then T^2 and its uncertainty:

$$T^2 = (2.35 \text{ s})^2 = 5.52 \text{ s}^2$$

$$\Delta T^2 = T^2 \left(\frac{2\Delta T}{T} \right) = 5.52 \text{ s}^2 \left(\frac{2 \times 0.03}{2.35} \right) = 0.14 \text{ s}^2$$

Thus $T^2 = 5.5 \pm 0.1 \text{ s}^2$

Remember, uncertainty is reported to 1 significant figure, and the value rounded to the same decimal place.

3 – Analysing data

3.1 General principles

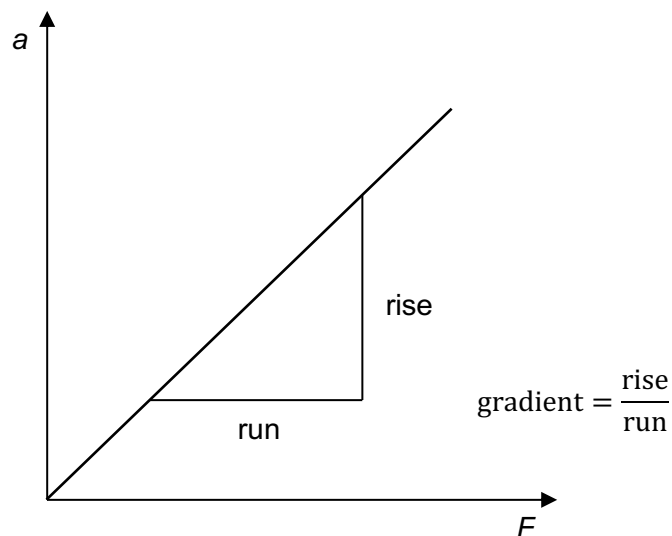
Data is often analysed by means of *linear regression*, i.e. finding the gradient of a straight-line graph. Straight lines are used because proportional relationships between variables are easy to identify.

Recall that the equation for a straight line can be given as

$$Y = AX + B$$

where A is the gradient of the line and B is the y -intercept.

Taking a simple example, say, Newton's second law $F = ma$, when the acceleration of an object was plotted against the force that was applied to it, a straight line would be obtained.



The equation could be linearized to:

$$a = \frac{1}{m} F$$
$$Y = AX + B$$

Diagram showing the correspondence between the variables in the equation $a = \frac{1}{m} F$ and the general linear equation $Y = AX + B$. Arrows indicate that a corresponds to Y , F corresponds to X , and $\frac{1}{m}$ corresponds to A .

where by correspondence, acceleration a is the y -variable and force F is the x -variable; the gradient is equal to $\frac{1}{m}$ or $\frac{1}{\text{mass}}$

The mass of the object could then be calculated by $\text{mass} = \frac{1}{\text{gradient}}$

Using a line of best fit to determine a relationship is preferable to simply substituting data pairs and finding unknowns algebraically because:

- measuring the gradient examines the relative changes in the variables, not the absolute values—this reduces systematic uncertainties.
- a gradient of a line of best fit is essentially an average of the ratio between the independent and dependent variables, reducing random uncertainties.

3.2 Linearized equations

Linear relationships are not always written in the form of the general straight line equation, and many other relationships in physics are not even linear. However, by careful manipulation of variables, many equations can be “made” to be linear. The table below shows some examples of how equations can be graphed to give a straight line.

The values for the y-variables and x-variables may need to be computed before being plotted on a graph. Values of interest can be computed by equating the gradient of the line with the terms in the “gradient” column.

Case	Equation	Linear form $Y = AX + B$	y-variable	x-variable	Gradient
The velocity of an object undergoing uniform acceleration is measured over time.	$v = u + at$	$v = at + u$	Final velocity v	Time t	Acceleration a
The acceleration of an object is measured when a varying amount of force is applied	$a = \frac{F}{m}$	$a = \frac{1}{m}F$	Acceleration a	Force F	Reciprocal of mass $\frac{1}{m}$
The period of a pendulum is measured when its length is varied	$T = 2\pi \sqrt{\frac{l}{g}}$	$T^2 = \frac{4\pi^2}{g}l$	Period squared T^2	Length l	$\frac{4\pi^2}{g}$
The vertical displacement of an object undergoing acceleration due to gravity	$\Delta s = -\frac{1}{2}gt^2$	$\Delta s = \left(-\frac{1}{2}g\right)t^2$	Change in vertical displacement Δs	Time-squared t^2	$-\frac{1}{2}g$
The intensity of light is measured at varying distances from its source	$I = I_0 \frac{1}{d^2}$	$I = I_0 \frac{1}{d^2}$	Intensity I	Reciprocal of distance squared $\frac{1}{d^2}$	Reference intensity I_0
The angle of refraction r is measured for every angle of incidence i	$n_1 \sin i = n_2 \sin r$	$\sin r = \frac{n_1}{n_2} \sin i$	$\sin r$	$\sin i$	Ratio of refractive indices $\frac{n_1}{n_2}$

3.3 Plotting graphs by hand

Good practice when hand drawing graphs in Stage 6 Physics includes graphs that:

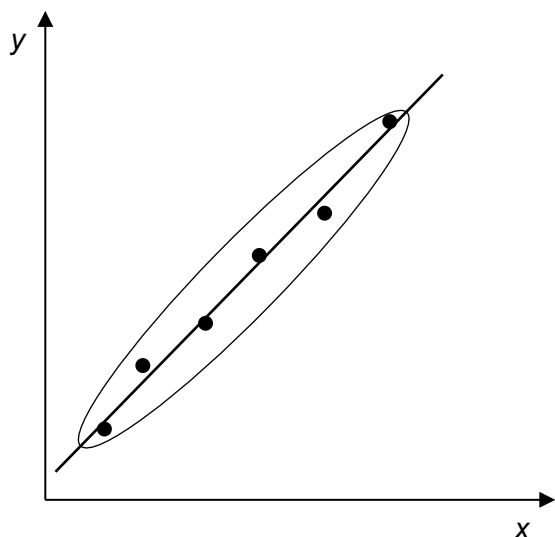
- are drawn with a lead pencil
- have axes labelled with appropriate labels and units
- usually have the independent variable on the x-axis
- usually have the dependent variable on the y-axis
- have graduations marked evenly along each axis
- provide a consistent linear scale which has been selected to allow the range of data displayed to extend over most of the available grid
- have a descriptive title or caption (a good formula is for this is “[dependent variable] versus [independent variable] for [context]”)
- plot points with small, sharp crosses (“x”)
- use a solid line of best fit that does not extend outside the range of plotted data
- have extrapolations outside the range of the plotted data as a dashed line
- display a large *rise-over-run* triangle drawn for determining a gradient.

3.3.1 Drawing lines of best fit

In Years 11 and 12, it is sufficient to estimate a line of best fit by eye. There are two approaches to drawing a line of best fit.

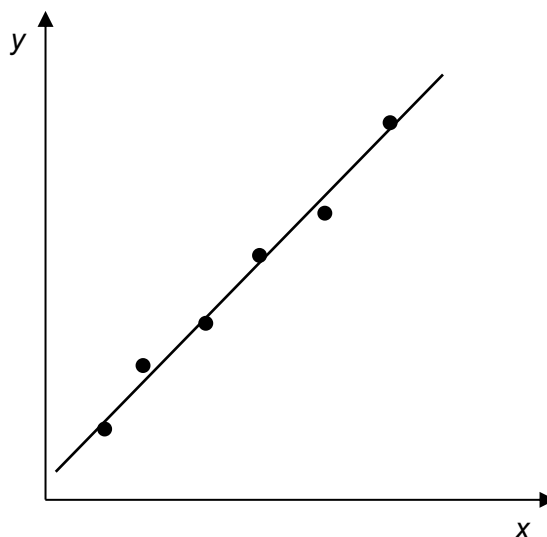
Using an ellipse

Draw an ellipse around all the data points, and then draw a line that bisects this ellipse, lengthways.



Just eyeballing it

Line up your ruler along the plotted data points and wiggle it around until you have a line that follows the trend down the middle. The data points should be roughly evenly distributed either side of the line.

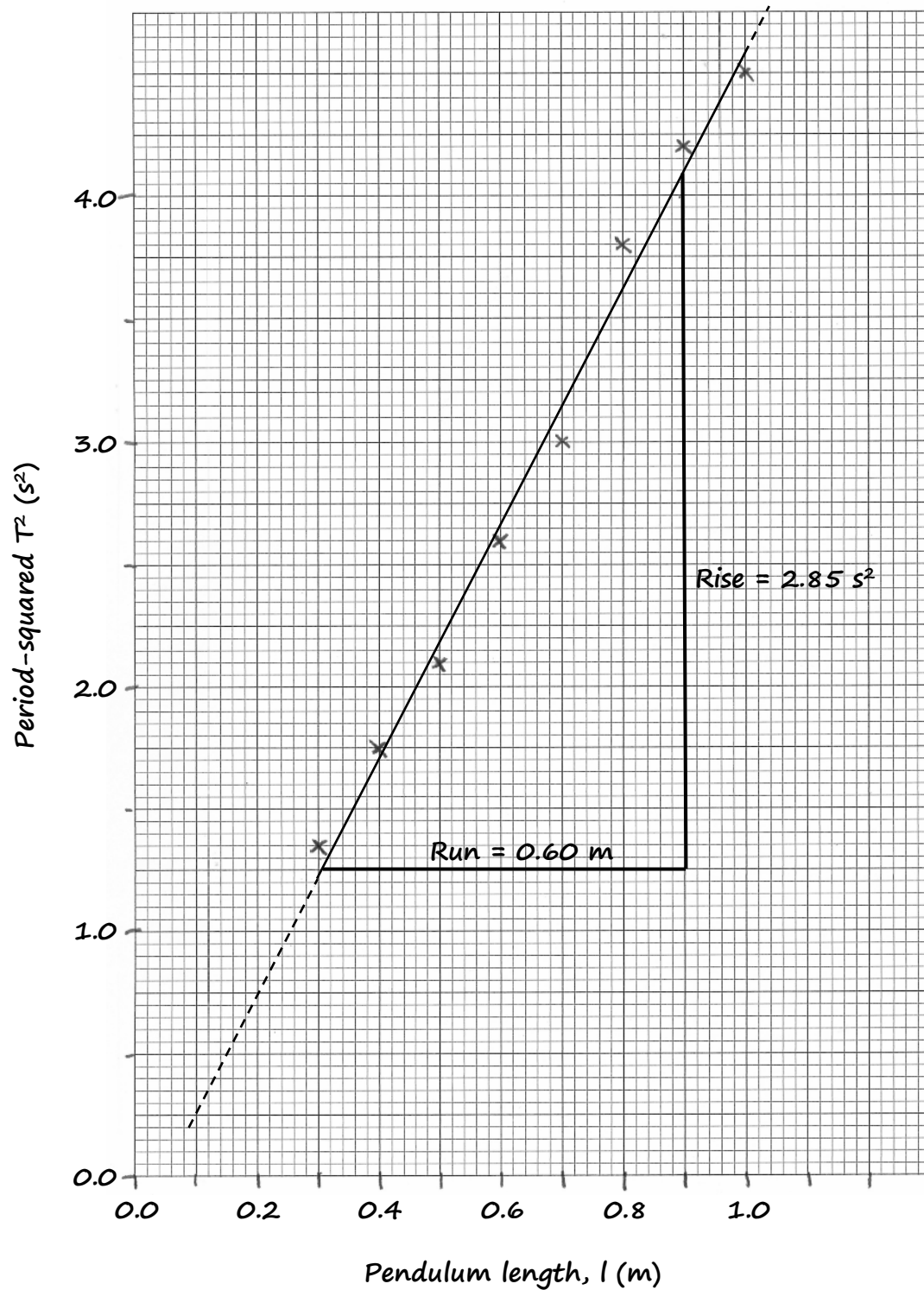


These two methods are fairly rough and ready. More accurate lines of best fit can be drawn using a spreadsheet (see the next section).

It is important to note that the line of best fit does not need to pass through any particular data points nor must it be forced to pass through the origin.

An example of a hand drawn graph.

Period-squared vs length for a pendulum



$$\text{Gradient} = \text{rise/run} = 2.85/0.60 = 4.75 \text{ s}^2 \text{ m}^{-1}$$

4 – Using spreadsheets

Spreadsheets are a powerful tool to process and analyse large amounts of data. You can try the following example, which uses *Google Sheets* (*Microsoft Excel* does the same thing in very similar ways).

4.1 Entering data

Click on a cell to select it and begin typing. Start by entering some column headings, remembering to include units. Enter your data next.

You can use the  button in the tool bar to put borders around your cells and lines in your table.

	A	B	C	D
1		10 Periods (s)		
2	Length (m)	Trial 1	Trial 2	Trial 3
3	0.30	11.25	11.7	11.76
4	0.40	13.56	13.05	13.06
5	0.50	14.34	14.3	14.53
6	0.60	16.35	15.93	16.23
7	0.70	17.23	17.04	17.74
8	0.80	19.63	19.41	19.24
9	0.90	20.47	20.73	20.19
10	1.00	21.2	20.33	22.01
11	1.10	21.15	22.12	22.09
12	1.20	23.33	23.92	23.3

4.2 Performing calculations

Next, you can perform calculations. In this example, we want to find the average of the three trials. It is important to note that spreadsheets locate data by the grid reference of the cell where it is contained.

At the top of a new column, put a heading (in this example, "Average"). In the same row as the first set of trials, type

`=AVERAGE(B3,C3,D3)`

And then press *Enter*.

	A	B	C	D	E	
1		10 Periods (s)				
2	Length (m)	Trial 1	Trial 2	Trial 3	Average	
3	0.30	11.25	11.7	11.7	<code>=AVERAGE(B3,C3,D3)</code>	
4	0.40	13.56	13.05	13.06		
5	0.50	14.34	14.3	14.53		
6	0.60	16.35	15.93	16.23		
7	0.70	17.23	17.04	17.74		
8	0.80	19.63	19.41	19.24		
9	0.90	20.47	20.73	20.19		
10	1.00	21.2	20.33	22.01		
11	1.10	21.15	22.12	22.09		
12	1.20	23.33	23.92	23.3		

The equals sign ("=") at the start of a formula tells the spreadsheet that the cell contains a calculation, and in this example, *B3*, *C3* and *D3* are the cells that we want to average. Hitting *Enter* after you type a formula in a cell commands the spreadsheet to execute your calculation.

Equivalently, we could have typed

`=AVERAGE(B3:D3)`

Using a colon as in “B3:D3” indicates a contiguous *range* of cells – could be useful if the number of cells you are calling on is large.

We do not want to keep typing a formula into the spreadsheet over and over again if we do not have to. Instead, we can just fill down a column with copies of the formula for its respective row.

Notice that there is a small square in the bottom right corner of a selected cell. When you hover your mouse pointer over this square, it turns into a crosshair. Click and drag this crosshair down to cover the rest of the empty cells in the *Average* column. When you release the mouse button, the formula fills into each cell and the averages for each row should automatically be computed.

	A	B	C	D	E
1		10 Periods (s)			
2	Length (m)	Trial 1	Trial 2	Trial 3	Average
3	0.30	11.25	11.7	11.76	11.57
4	0.40	13.56	13.05	13.06	
5	0.50	14.34	14.3	14.53	
6	0.60	16.35	15.93	16.23	
7	0.70	17.23	17.04	17.74	
8	0.80	19.63	19.41	19.24	
9	0.90	20.47	20.73	20.19	
10	1.00	21.2	20.33	22.01	
11	1.10	21.15	22.12	22.09	
12	1.20	23.33	23.92	23.3	

Click, drag then release

E
Average
11.57
13.22333333
14.39
16.17
17.33666667
19.42666667
20.46333333
21.18
21.78666667
23.51666667

Your *Average* column may look like shown above, right. To reduce the number of decimal places, click on the  button in the toolbar while the cells are selected until there is an appropriate number of decimal places.

Now, in this example, we want to run calculations to find the time for *one period* and also *period squared*. To do this, we will need to type some formulae into cells F3 and G3:

- We want to compute a value for *one period* in cell F3. Since cell E3 contains the time for *ten periods* we will divide this by 10. Type into cell F3:

`=E3/10`

- We want to compute a value for *period-squared* in cell G3. Since cell F3 now contains the time for *one period*, type into cell G3:

`=F3^2`

Remember to fill the formulae down the columns as before. If there are too many decimal places, use the  button again while the appropriate cells are selected.

The completed table should look like the one below.

	A	B	C	D	E	F	G
1		10 Periods (s)					
2	Length (m)	Trial 1	Trial 2	Trial 3	Average	Period (s)	Period^2 (s^2)
3	0.30	11.25	11.7	11.76	11.57	1.16	1.34
4	0.40	13.56	13.05	13.06	13.22	1.32	1.75
5	0.50	14.34	14.3	14.53	14.39	1.44	2.07
6	0.60	16.35	15.93	16.23	16.17	1.62	2.61
7	0.70	17.23	17.04	17.74	17.34	1.73	3.01
8	0.80	19.63	19.41	19.24	19.43	1.94	3.77
9	0.90	20.47	20.73	20.19	20.46	2.05	4.19
10	1.00	21.2	20.33	22.01	21.18	2.12	4.49
11	1.10	21.15	22.12	22.09	21.79	2.18	4.75
12	1.20	23.33	23.92	23.3	23.52	2.35	5.53

4.3 Plotting a graph

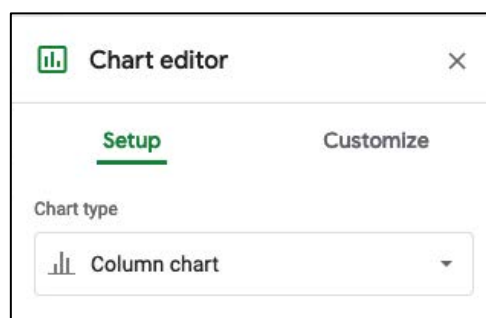
Select the cells in the columns of your table that you wish to use for your graph. Click in the centre of the first cell (including the column header) and drag down until the cells are selected, and then release the mouse button.

In this example, the columns are not adjacent so they cannot be selected in one step. To select column cells that are not adjacent, you will need to click and drag to select the first column (as above), release, then holding down the *ctrl* key, click, drag and release when selecting the cells of the other column.

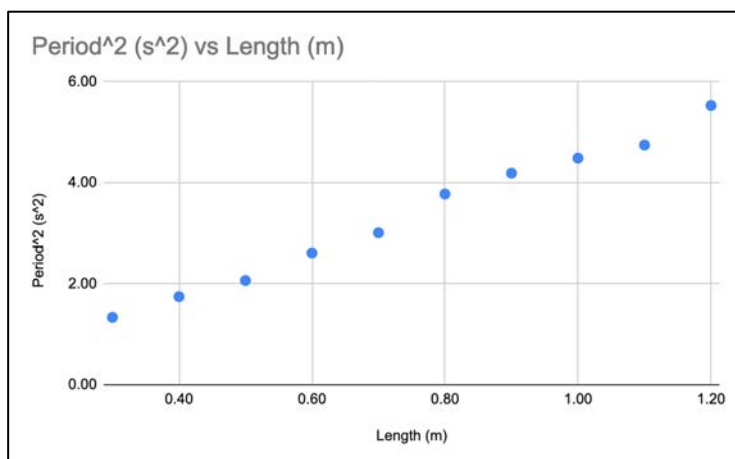
	A	B	C	D	E	F	G
1		10 Periods (s)					
2	Length (m)	Trial 1	Trial 2	Trial 3	Average	Period (s)	Period^2 (s^2)
3	0.30	11.25	11.7	11.76	11.57	1.16	1.34
4	0.40	13.56	13.05	13.06	13.22	1.32	1.75
5	0.50	14.34	14.3	14.53	14.39	1.44	2.07
6	0.60	16.35	15.93	16.23	16.17	1.62	2.61
7	0.70	17.23	17.04	17.74	17.34	1.73	3.01
8	0.80	19.63	19.41	19.24	19.43	1.94	3.77
9	0.90	20.47	20.73	20.19	20.46	2.05	4.19
10	1.00	21.2	20.33	22.01	21.18	2.12	4.49
11	1.10	21.15	22.12	22.09	21.79	2.18	4.75
12	1.20	23.33	23.92	23.3	23.52	2.35	5.53

With the cells selected, click on the  button in the toolbar, or alternatively, select **Insert > Chart** from the menu. A graph should pop up on the screen and a *Chart editor* pane will open on the right.

By default, *Google Sheets* will usually generate a column graph. To change this to a scatter graph, use the *Chart type* dropdown menu under the *Setup* tab in the *Chart Editor* and select *Scatter*.

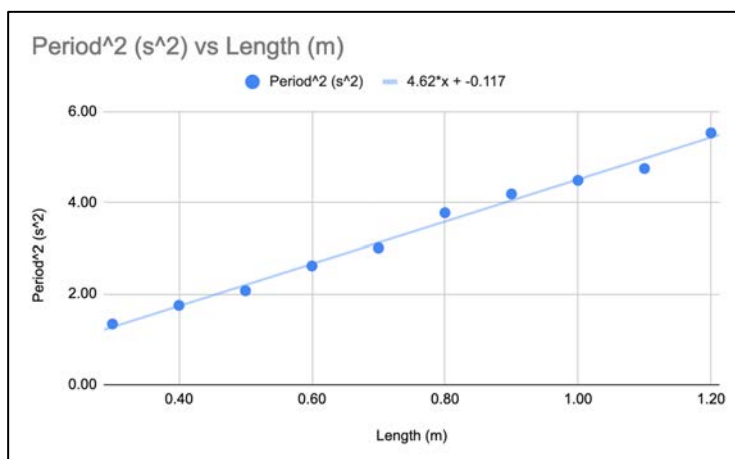


If your dependent variable is in the left column of the table and the independent variable is in the right column, then the dependent and independent variables should be positioned and labelled on the appropriate axes.



To add a line of best fit, expand *Series* under the *Customize* tab in the *Chart Editor* pane (see right). Tick the *Trendline* box. We also want the equation of the line – from the *Label* dropdown menu, select *Use Equation*.

Your graph should now look like the one below.



Notice that the equation of the line is now shown above the graph area; the gradient is 4.62 and the y-intercept is -0.117.

Chart editor

Setup

Customize

Series

Period^2 (s^2)

Format

Color

Axis

Point size

Point shape

Format data point

Error bars

Data labels

Trendline

Type

Line opacity

Label

Line color

Line thickness

Show R^2

4.4 Obtaining the uncertainty in the slope

Remember that in section 2.2.4 *Uncertainty in the slope of a line of best fit*, we saw that there is inherent uncertainty in the calculated gradient of a line of best fit.

There is a function in *Google Sheets* (`=LINEST(Y_VALUES, X_VALUES, TRUE, TRUE)`) that easily allows you to compute the uncertainty in the line (the function is the same in *Microsoft Excel*, but it's a bit trickier to get *Excel* to display the result). It should be noted that this method does not take the uncertainty in each data point into consideration – it is looking at the spread of datapoints from the trend line.

Using our example, choose a cell below or to the right of your data (make sure this cell is clear for 5 rows down and another column right – it will fill these cells with data). Into this cell, type:

`=LINEST(G3:G12, A3:A12, TRUE, TRUE)`

where G3:G12 is the range of y-values and A3:A12 is the range of x-values.

	A	B	C	D	E	F	G
1		10 Periods (s)					
2	Length (m)	Trial 1	Trial 2	Trial 3	Average	Period (s)	Period^2 (s^2)
3	0.30	11.25	11.7	11.76	11.57	1.16	1.34
4	0.40	13.56	13.05	13.06	13.22	1.32	1.75
5	0.50	14.34	14.3	14.53	14.39	1.44	2.07
6	0.60	16.35	15.93	16.23	16.17	1.62	2.61
7	0.70	17.23	17.04	17.74	17.34	1.73	3.01
8	0.80	19.63	19.41	19.24	19.43	1.94	3.77
9	0.90	20.47	20.73	20.19	20.46	2.05	4.19
10	1.00	21.2	20.33	22.01	21.18	2.12	4.49
11	1.10	21.15	22.12	22.09	21.79	2.18	4.75
12	1.20	23.33	23.92	23.3	23.52	2.35	5.53
13							
14							
15	<code>=LINEST(G3:G12, A3:A12, TRUE, TRUE)</code>						
16							

When you press *Enter*, the function will output this array of data:

Gradient	4.622671939	-0.1167532545	y-intercept
Uncertainty	0.1518706562	0.1219701975	
	0.9914391352	0.1379433738	
	926.4850336	8	
	17.62950408	0.1522269951	

The ones we are interested in are the gradient, y-intercept, and uncertainty in the slope. The gradient can be quoted as 4.6 ± 0.2 .

Notice that the gradient and y-intercept here are given to far more significant figures than that labelled on the graph.

4.5 Common spreadsheet symbols and functions

4.5.1 Calling on cells in functions

Data in a cell can be called upon in a formula by their grid coordinate, for example, A1, B4, H27, and so on.

If a contiguous range of adjacent cells in a row, column, or block is being called upon, then a colon ":" can be used. For example:

- B4:B9 would call upon six cells in column B of the spreadsheet.
- G2:K2 would call upon five cells in row 2.
- B2:D5 would call on twelve cells in a rectangle between columns B to D and rows 2 to 5.

This is particularly useful when computing the average of a large number of cells, for example, =AVERAGE(D2:D99).

4.5.2 Orders of operations

Use brackets to specify order of operations. It is better to be over cautious and use more brackets than fewer if you are unsure about the order in which the spreadsheet will compute a formula. For example:

An equation that looks like this $a = \frac{v - u}{t}$

would need to be entered with brackets like this $=(B-A)/C$

4.5.3 Entering data

Below are some special inputs for a spreadsheet,

Case	Spreadsheet	Example (natural)	Example syntax
Pi, π	pi()	$2\pi r$	2*pi()*A1
Euler's number	exp(A)	e^2	exp(2)
Scientific notation	E	5.23×10^3	5.23E+3

4.5.4 Common functions

Below is a table of some commonly used spreadsheet functions, where A, B et cetera might be numbers or a cell references. This is by no means an exhaustive list of spreadsheet functions.

Some functions use an argument placed inside brackets, such as for square root, $SQRT(A)$. The argument can be a number, a reference to a cell containing a number, or it can even be another function. For example, $\sqrt{5x}$ can be entered as $SQRT(5*A1)$.

Remember to begin any calculation with an equal sign “=”.

Function	Example	Syntax
Addition	$2 + x$	=A+B
Subtraction	$x - 3$	=A-B
Multiplication	$7 \times x$	=A*B
Division	$\frac{x}{4}$	=A/B
Indices	x^3	=A^B
Square root	$\sqrt{x}$	=SQRT(A)
Logarithm	$\log_b a$, e.g. $\log_{10} x$	=log(num,base)
Sine*	$\sin x$	=SIN(A) =SIN(RADIANS(A))
Cosine*	$\cos x$	=COS(A)
Tangent*	$\tan x$	=TAN(A)
Arcsine*	$\sin^{-1} x$	=ASIN(A) =DEGREES(ASIN(A))
Arccosine*	$\cos^{-1} x$	=ACOS(A)
Arctangent*	$\tan^{-1} x$	=ATAN(A)
Degrees to radians	$\frac{x}{180} \times \pi$	=RADIANS(A)
Radians to degrees	$\frac{x}{\pi} \times 180^\circ$	=DEGREES(A)
Average	$\frac{\sum_1^n x}{n}$	=AVERAGE(A, B, C, D) =AVERAGE(A:D)

*Spreadsheets input and output angles in radians.

Exercises 1

Recording data

1. Write the following numbers in scientific notation.

4860 J

0.00761 m

84 kg

2. Write the following numbers in natural notation.

5.39×10^3 A

7.22×10^{-6} W

9.3×10^0 Pa

3. What orders of magnitude are these numbers?

4.734×10^7 V

6.54×10^{-5} kg

3.6155×10^3 W

4. How many significant figures are in these numbers?

15400 N

0.0007054 s

248.140 K

5. Round these numbers to three significant figures.

256 190 W

4.389 L

16.347 Ω

6. Write these observations in their base SI unit.

378 μm

0.0391 MJ

6.71×10^6 V

7. Write these observations in more convenient units.

6 428 000 V

0.0000088 W

101 800 Pa

8. Write the results of these calculations with the correct units and significant figures.

$$\text{Impulse} = 16.350 \text{ N} \times 9.87 \text{ s}$$

$$\text{Spring constant, } k = \frac{6.89 \text{ N}}{0.025 \text{ m}}$$

$$\text{Refractive index} = \frac{3.0 \times 10^8 \text{ ms}^{-1}}{2.54 \times 10^8 \text{ ms}^{-1}}$$

Exercises 2

Uncertainties

1. The accepted value for the specific heat capacity of water is accepted to be $4186 \text{ J kg}^{-1} \text{ K}^{-1}$. In an investigation to measure it experimentally, a student produces a value of $4390 \text{ J kg}^{-1} \text{ K}^{-1}$.

What is:

- (a) The absolute error?
- (b) The relative error (as a percentage)?
- (c) The accuracy?

2. Calculate the relative uncertainties for the following values:

$$57.9 \pm 0.3 \text{ m}$$

$$1.7 \pm 0.2 \text{ kg}$$

$$2510 \pm 20 \text{ years}$$

3. Look at the following measuring devices. What uncertainty can be expected from observations using each?

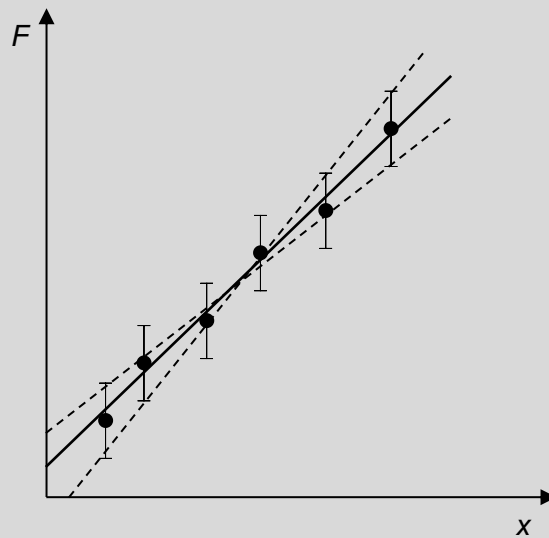


4. Consider the following repeated measurements of temperature:

$$19.6^{\circ}\text{C}, 19.4^{\circ}\text{C}, 19.2^{\circ}\text{C}, 19.5^{\circ}\text{C}, 19.8^{\circ}\text{C}$$

Calculate the average and its uncertainty.

5. Consider the following graph of the magnitude of the restoring force of a spring F versus the extension of a spring, x . It is known that F and x are related by Hooke's law, $F = kx$, where k is a spring constant.



The gradients of each line were computed:

- Line of best fit: 4.7 N m^{-1}
- Line of worst fit (max): 4.9 N m^{-1}
- Line of worst fit (min): 4.4 N m^{-1}

Calculate the value for the spring constant k and its uncertainty.

6. Calculate the results and uncertainties of the following:

$$42.0 \pm 0.2 \text{ g} + 27.2 \pm 0.2 \text{ g}$$

$$6.3 \pm 0.1 \text{ m} - 4.9 \pm 0.1 \text{ m}$$

$$2 \times 78.2 \pm 0.5 \text{ g}$$

$$23.1 \pm 0.5 \text{ W} \times 11.2 \pm 0.1 \text{ s}$$

$$2.45 \pm 0.05 \text{ N} \div 0.250 \pm 0.010 \text{ kg}$$

$$(7.5 \pm 0.3 \text{ m s}^{-1})^2$$

Exercises 3

Analysing data

1. Rewrite the following equations into the linear form $Y = AX + B$ given the dependent and independent variables, and then identify the terms of the gradient.

- a) For the heat required to boil varying masses of water

$$Q = mc\Delta T$$

Independent variable – mass of water m

Dependent variable – heat, Q

Controlled variables – specific heat c and change in temperature ΔT (room temp to boiling)

- b) For an object undergoing uniform circular motion

$$F = \frac{mv^2}{r}$$

Independent variable – radius r

Dependent variable – Centripetal force, F

Controlled variables – mass m and linear speed v

2. Use the data in this table to draw a graph of T^2 versus l . Determine its gradient and hence g .

Length l (m)	T^2 (s ²)
0.30	1.34
0.40	1.75
0.50	2.07
0.60	2.61
0.70	3.00
0.80	3.77
0.90	4.19
1.00	4.49
1.10	4.75
1.20	5.53

Solutions to Exercises

Recording data

1. 4.86×10^3 J, 7.61×10^{-3} m, 8.4×10^1 kg
2. 5390 A, 0.00000722 W, 9.3 Pa
3. 7, -4 (because 6.54×10^{-5} rounds up to 1×10^{-4}), 3
4. Three significant figures, four significant figures, six significant figures
5. 256 000 W, 4.39 L, 16.3 Ω
6. 3.78×10^{-6} m, 3.91×10^4 J, 6 710 000 V
7. 6.428 MV, 8.8 μ W, 1018 hPa (meteorologists often quote atmospheric pressures in hectopascals! Otherwise 101.8 kPa is acceptable.)
8. 161 Ns, 280 Nm^{-1} , 1.2 (no units!)

Uncertainties

1. (a) $204 \text{ J kg}^{-1} \text{ K}^{-1}$ (b) 4.9% (c) 95.1%
2. 0.5%, 12%, 0.8%
3. Ruler ± 0.05 cm, stopwatch ± 0.005 s, protractor $\pm 0.5^\circ$, balance ± 0.05 g, digital multimeter in 10ADC mode ± 0.005 A
4. $19.5 \pm 0.3^\circ \text{C}$
5. $4.7 \pm 0.3 \text{ N m}^{-1}$
6. 69.2 ± 0.4 g, 1.4 ± 0.2 m, 156.4 ± 1.0 g, 259 ± 8 Ws, $9.8 \pm 0.6 \text{ N kg}^{-1}$, $56 \pm 5 \text{ m}^2 \text{s}^{-2}$

Analysing data

1. (a) $Q = c\Delta T \cdot m$, gradient = $c\Delta T$ (b) $F = mv^2/1/r$, gradient = mv^2 (F would be graphed against $1/r$)
2. See example on page 15.

References

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First Year Physics Unit 2020, *First year laboratory manual, Physics 1A*, School of Physics, UNSW Sydney.

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																				
KEY																				
		Atomic Number Symbol Standard Atomic Weight Name																		
		79 Au 197.0 Gold																		
1 H 1.008 Hydrogen	4 Be 9.012 Beryllium	12 Mg 24.31 Magnesium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton	
3 Li 6.941 Lithium	11 Na 22.99 Sodium	19 K 39.10 Potassium	37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc 101.1 Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	87 Fr	56 Ba 137.3 Barium	57-71 Lanthanoids	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po 209.0 Polonium	85 At 209.0 Astatine	86 Rn 209.0 Radon	87 Fr	
87 Fr	88 Ra	89-103 Actinoids	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og	119 Tennessine	120 Oganesson	

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm 144.2 Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu 244.0 Plutonium	95 Am 243.0 Americium	96 Cm 247.0 Curium	97 Bk 247.0 Berkelium	98 Cf 251.0 Californium	99 Es 252.0 Einsteinium	100 Fm 257.0 Fermium	101 Md 288.1 Mendelevium	102 No 289.1 Nobelium	103 Lr 260.1 Lawrencium
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Standard atomic weights are abridged to four significant figures. Elements with no reported values in the table have no stable nuclides. Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.